Finger Pose Classification — Run Report

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Data paths

Data root	/home/jestin/ThesisRepo/ML/NewTrainingData/Finger_MCP_PIP · 241 trials total
Class folders	curl_curl, curl_straight, straight_curl, straight_straight
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/FingerPoseReports
Trials · curl_curl	61
Trials · curl_straight	60
Trials · straight_curl	60
Trials · straight_straight	60

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky
Modalities	YPR, Accel, Flex
Sensor columns	164 channels
Fingers	thumb, index, middle, ring, pinky
Classes	4: curl_curl, curl_straight, straight_curl, straight_straight

Preprocessing, features & split

Resample steps	90
Butterworth	on · cutoff 5.0 Hz · order 4 · fs 30.0 Hz
Normalisation	standard
Feature mode	stats → feature dim 160
Split / seed / CV	test_size=0.2 · random_state=42 · CV folds=5
Trials per finger dataset	817 (train+test, post-aug train)

Augmentation (training only)

Copies per train trial	3
Gaussian noise	on · sigma=0.02 (per-channel std-scaled)
Time warp	on · max_scale=0.1
Amplitude scaling	off · range=(0.9, 1.1)

Active classifiers

Algorithm	Class
SVM (RBF)	SVC
SVM (Linear)	SVC
Random Forest	RandomForestClassifier
KNN	KNeighborsClassifier
Logistic Regression	LogisticRegression
LDA	LinearDiscriminantAnalysis
Voting Soft (SVM+RF+KNN+LogReg)	VotingClassifier
Voting Soft (SVM+RF)	VotingClassifier

Algorithm	Class
Stacking (SVM+RF+KNN+LogReg)	StackingClassifier

Classifier hyperparameters

SVM (RBF)	C=10 · break_ties=False · cache_size=200 · class_weight=None · coef0=0.0 · decision_function_shape=ovr · degree=3 · gamma=scale · kernel=rbf · max_iter=-1 · probability=True · random_state=42 · shrinking=True · tol=0.001 · verbose=False
SVM (Linear)	C=1 · break_ties=False · cache_size=200 · class_weight=None · coef0=0.0 · decision_function_shape=ovr · degree=3 · gamma=scale · kernel=linear · max_iter=-1 · probability=True · random_state=42 · shrinking=True · tol=0.001 · verbose=False
Random Forest	bootstrap=True · ccp_alpha=0.0 · class_weight=None · criterion=gini · max_depth=None · max_features=sqrt · max_leaf_nodes=None · max_samples=None · min_impurity_decrease=0.0 · min_samples_leaf=1 · min_samples_split=2 · min_weight_fraction_leaf=0.0 · monotonic_cst=None · n_jobs=-1 · oob_score=False · random_state=42 · verbose=0 · warm_start=False
KNN	algorithm=auto · leaf_size=30 · metric=euclidean · metric_params=None · n_jobs=None · n_neighbors=5 · p=2 · weights=uniform
Logistic Regression	C=1.0 · class_weight=None · dual=False · fit_intercept=True · intercept_scaling=1 · l1_ratio=0.0 · max_iter=1000 · n_jobs=None · penalty=deprecated · random_state=42 · solver=lbfgs · tol=0.0001 · verbose=0 · warm_start=False
LDA	covariance_estimator=None · n_components=None · priors=None · shrinkage=None · solver=svd · store_covariance=False · tol=0.0001
Voting Soft (SVM+RF+KNN+LogReg)	flatten_transform=True · n_jobs=-1 · verbose=False · voting=soft · weights=None
Voting Soft (SVM+RF)	flatten_transform=True · n_jobs=-1 · verbose=False · voting=soft · weights=None
Stacking (SVM+RF+KNN+LogReg)	cv=5 · final_estimator=LogisticRegression(max_iter=1000, random_state=42) · n_jobs=-1 · passthrough=False · stack_method=auto · verbose=0

Per-dataset test accuracy

Rows = (hand, finger) dataset. Columns = classifier. Last row = mean across datasets.

Dataset	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
left · thumb	0.755	0.755	0.857	0.673	0.673	0.653	0.776	0.776	0.735
left · index	0.755	0.714	0.857	0.776	0.694	0.714	0.816	0.816	0.796
left · middle	0.816	0.755	0.959	0.776	0.796	0.776	0.837	0.837	0.816
left · ring	0.857	0.776	0.980	0.714	0.816	0.898	0.878	0.878	0.878
left · pinky	0.776	0.878	0.898	0.653	0.837	0.755	0.837	0.837	0.816
right · thumb	0.918	0.959	0.980	0.571	0.939	0.918	0.918	0.918	0.959
right · index	0.837	0.898	0.980	0.776	0.878	0.878	0.898	0.898	0.898
right · middle	0.878	0.796	0.959	0.735	0.776	0.796	0.837	0.837	0.837
right · ring	0.837	0.796	0.898	0.653	0.776	0.776	0.796	0.796	0.857
right · pinky	0.918	0.959	0.939	0.796	0.939	0.837	0.980	0.980	0.980
Mean	0.835	0.829	0.931	0.712	0.812	0.800	0.857	0.857	0.857

Per-class accuracy by dataset

Each table: rows = class, columns = classifier. Values are per-class accuracy on the test set.

left · thumb

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.769	0.923	0.769	0.769	0.692	0.923	0.923	0.846
curl_straight	0.917	0.833	0.917	0.917	0.833	0.833	0.917	0.917	0.917
straight_curl	0.667	0.833	1.000	0.500	0.583	0.417	0.750	0.750	0.667
straight_straight	0.500	0.583	0.583	0.500	0.500	0.667	0.500	0.500	0.500

left · index

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	1.000	0.923	0.923	0.923	0.923	0.769	0.923	0.923	0.923
curl_straight	0.917	0.917	1.000	0.917	0.917	0.833	1.000	1.000	1.000
straight_curl	0.417	0.417	0.833	0.417	0.417	0.583	0.583	0.583	0.500
straight_straight	0.667	0.583	0.667	0.833	0.500	0.667	0.750	0.750	0.750

left · middle

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.846	0.923	0.846	0.846	0.923	0.923	0.923	0.846

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_straight	0.833	0.833	1.000	0.917	0.917	0.750	0.917	0.917	0.917
straight_curl	0.833	0.750	0.917	0.750	0.750	0.583	0.833	0.833	0.833
straight_straight	0.667	0.583	1.000	0.583	0.667	0.833	0.667	0.667	0.667

left · ring

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.769	0.923	0.846	0.692	0.846	0.923	0.923	0.923
curl_straight	0.917	0.917	1.000	0.917	0.917	0.917	0.917	0.917	0.917
straight_curl	0.833	0.750	1.000	0.500	0.917	1.000	0.917	0.917	0.917
straight_straight	0.750	0.667	1.000	0.583	0.750	0.833	0.750	0.750	0.750

left · pinky

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.846	0.846	0.923	0.846	0.846	0.769	0.846	0.846	0.846
curl_straight	0.833	1.000	0.917	0.583	1.000	0.833	0.917	0.917	0.917
straight_curl	0.583	0.750	0.833	0.500	0.583	0.583	0.667	0.667	0.583
straight_straight	0.833	0.917	0.917	0.667	0.917	0.833	0.917	0.917	0.917

right · thumb

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.923	0.923	0.615	0.923	0.923	0.923	0.923	0.923
curl_straight	1.000	1.000	1.000	0.500	1.000	0.917	1.000	1.000	1.000
straight_curl	1.000	0.917	1.000	0.750	0.917	0.917	0.917	0.917	0.917
straight_straight	0.750	1.000	1.000	0.417	0.917	0.917	0.833	0.833	1.000

right · index

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.923	0.923	0.846	0.923	0.923	0.923	0.923	0.923
curl_straight	0.917	1.000	1.000	0.667	1.000	1.000	1.000	1.000	1.000
straight_curl	0.833	0.833	1.000	0.833	0.750	0.833	0.833	0.833	0.833
straight_straight	0.667	0.833	1.000	0.750	0.833	0.750	0.833	0.833	0.833

right · middle

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.923	0.923	0.923	0.923	0.923	0.923	0.923	0.923
curl_straight	0.833	0.833	1.000	0.750	0.833	0.833	0.833	0.833	0.917
straight_curl	1.000	0.750	1.000	0.667	0.750	0.833	0.917	0.917	0.833
straight_straight	0.750	0.667	0.917	0.583	0.583	0.583	0.667	0.667	0.667

right · ring

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.923	0.923	0.846	0.923	0.846	0.923	0.923	0.923
curl_straight	0.750	0.833	0.833	0.667	0.833	0.833	0.833	0.833	0.833
straight_curl	0.917	0.750	1.000	0.417	0.750	0.750	0.750	0.750	0.917
straight_straight	0.750	0.667	0.833	0.667	0.583	0.667	0.667	0.667	0.750

right · pinky

Finger	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Voting Soft (SVM+RF)	Stacking (SVM+RF+KNN+LogReg)
curl_curl	0.923	0.923	0.846	0.769	0.923	0.769	0.923	0.923	0.923
curl_straight	1.000	1.000	1.000	0.750	0.917	1.000	1.000	1.000	1.000
straight_curl	0.917	0.917	0.917	0.750	1.000	0.833	1.000	1.000	1.000
straight_straight	0.833	1.000	1.000	0.917	0.917	0.750	1.000	1.000	1.000